Example 4.13 Find the total number of outcomes when 5 coins are tossed once.

Solution:
When a coin is tossed, the outcomes are in two ways which are $\{Head, Tail\}$.
By the rule of product, the number of outcomes when 5 coins are tossed is $2 \times 2 \times 2 \times 2 \times 2 = 2^5 = 32$.



More generally, if n coins are tossed then the number of outcomes is 2^n.

Example 4.14 In how many ways (i) 5 different balls be distributed among 3 boxes? (ii) 3 different balls be distributed among 5 boxes?

Solution:

(i) Each ball can be placed into any one of the three boxes in 3 different ways. Therefore, by rule of product, the number of ways of distributing 5 different balls among three boxes is $3 \times 3 \times 3 \times 3 \times 3 = 3^5 = 243$.
(ii) Each ball can be placed into any one of the five boxes in 5 different ways. Therefore, by rule of product, the number of ways of distributing 3 different balls among five boxes is $5 \times 5 \times 5 = 5^3 = 125$.



In order to avoid confusions, take the objects(balls) and distribute them in places(boxes). More generally, if n different objects are to be placed in m places, then the number of ways of placing is m^n.

Example 4.15 There are 10 bulbs in a room. Each one of them can be operated independently. Find the number of ways in which the room can be illuminated.

Solution:
Each of the 10 bulbs are operated independently means that each bulb can be operated in two ways. That is in off mode or on mode. The total number of doing this are 2^{10} which includes the case in which 10 bulbs are off. Keeping all 10 bulbs in "off" mode, the room cannot be illuminated. Hence, the total number of ways the room can be illuminated are $2^{10} - 1 = 1024 - 1 = 1023$.

Another concept which is an essential tool in a counting process which is stated as follows:

The Pigeonhole Principle:

Suppose a flock of pigeons fly into a set of pigeonholes. If there are more pigeons than pigeonholes then there must be at least one pigeonhole with at least two pigeons in it. A generalised form of this may be applied to other objects and situations as well.
If $k + 1$ or more objects are placed in k boxes, then there is at least one box containing two or more of the objects.
Here are some examples.

1. In any group of 27 English words, there must be at least two words which begin with the same letter (since there are only 26 letters in the English alphabet).
2. If six meetings are held 5 working days of a week, then there must be at least two meetings held on the same day.